

TechNova Marketplace – Complete Project Documentation

Project Name

TechNova Marketplace

Tagline

A Modern Full-Stack Electronics E-Commerce Platform for Smart Shopping

Document 01 — Product Requirements Document (PRD)

Project Overview

TechNova Marketplace is a full-stack electronics e-commerce platform that allows customers to browse, search, compare, and purchase electronic products online.

The platform provides a seamless shopping experience while enabling administrators to manage inventory, orders, customers, and product catalogs efficiently.

Problem Statement

Many small electronics retailers lack professional online stores.

Common issues include:

- Poor user experience
- Manual inventory management
- Lack of order tracking
- No centralized administration
- Slow product discovery

TechNova Marketplace solves these challenges through a scalable web-based platform.

Target Users

Primary Users

Customers interested in purchasing:

- Laptops
- Smartphones
- Accessories
- Smart Devices
- Gaming Products

Secondary Users

Store Administrators

Responsibilities:

- Product Management
 - Inventory Management
 - Order Management
 - Customer Monitoring
-

Core Features

Authentication

- User Registration
- Login
- JWT Authentication
- Password Encryption
- Role-Based Authorization

Product Catalog

- Product Listing
- Product Details
- Product Categories
- Product Search
- Product Filters
- Product Sorting

Shopping Features

- Add to Cart
- Update Quantity
- Remove From Cart
- Checkout
- Order Placement
- Order Tracking

User Dashboard

- Profile Management
- Order History
- Saved Addresses

Admin Dashboard

- Dashboard Analytics
 - Product Management
 - Inventory Control
 - User Management
 - Order Management
-

Nice-To-Have Features

- Wishlist
 - Product Reviews
 - Ratings
 - Coupon System
 - Product Comparison
 - Recently Viewed Products
 - Email Notifications
-

Out Of Scope

Version 1 will NOT include:

- AI Recommendations
 - Live Chat
 - Multi Vendor Marketplace
 - Mobile App
 - Cryptocurrency Payments
-

User Stories

Customer

As a customer, I want to search products so I can quickly find items.

As a customer, I want to place orders securely.

As a customer, I want to track order status.

As a customer, I want to view previous purchases.

Admin

As an admin, I want to manage products.

As an admin, I want to monitor orders.

As an admin, I want to update stock.

Success Metrics

Technical

- API Response < 2 Seconds
- Secure JWT Authentication
- Responsive Design

Business

- Product Conversion Rate
 - Order Completion Rate
 - User Retention
-

Functional Requirements

Frontend

- React UI
- Tailwind Styling
- Axios Integration
- Dynamic Components

Backend

- REST APIs
- JWT Security
- CRUD Operations

Database

- MySQL Relational Database
 - Indexed Queries
 - Foreign Key Relationships
-

Document 02 — Technical Requirements Document (TRD)

Frontend Stack

- React JS
- Tailwind CSS
- React Router DOM
- Axios

Backend Stack

- Node JS
- Express JS

Database

- MySQL

Authentication

- JWT
- bcryptjs

Development Tools

- VS Code
- Git
- GitHub
- Postman
- XAMPP

Folder Structure

frontend/

- components/
- pages/
- layouts/
- routes/
- hooks/
- context/
- services/

backend/

- controllers/
- models/
- routes/
- middleware/
- config/
- utils/

Environment Variables

PORT=

DB_HOST=

DB_USER=

DB_PASSWORD=

DB_NAME=

JWT_SECRET=

Document 03 — App Flow

Guest User

Home

↓

Products

↓

Product Details

↓

Login/Register

↓

Cart

Customer Journey

Home



Products



Product Details



Add To Cart



Cart



Checkout



Order Confirmation

Admin Journey

Admin Login



Dashboard



Manage Products



Manage Orders



Inventory

Routes

/

/login

/register

/products

/product/:id

/cart

/checkout

/orders

/profile

/admin

/admin/products

/admin/orders

/admin/users

Document 04 — UI/UX Design Brief

Design Theme

Modern Technology Store

Inspired By:

- Apple
 - Samsung
 - ASUS
 - Dell
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Color Palette

Primary

0F172A

Secondary

2563EB

Accent

3B82F6

Background

F8FAFC

Cards

FFFFFF

Text

1E293B

Typography

Font:

Poppins

Heading:

36px

Section:

28px

Body:

16px

Components

Navbar

- Logo
- Search Bar
- Categories
- Cart
- User Menu

Product Cards

- Product Image
- Product Name
- Price
- Rating
- Add To Cart

Dashboard

- Statistics Cards
 - Tables
 - Sidebar Navigation
-

Document 05 — Database Schema

users

id

name

email

password

role

created_at

categories

id

name

slug

products

id

category_id

title

description

price

brand

stock

image_url

created_at

cart

id

user_id

product_id

quantity

orders

id

user_id

total_amount

status

created_at

order_items

id

order_id

product_id

quantity

price

addresses

id

user_id

full_name

phone

city

state

pincode

address

Relationships

users.id → orders.user_id

users.id → cart.user_id

products.id → cart.product_id

orders.id → order_items.order_id

products.id → order_items.product_id

categories.id → products.category_id

API Endpoints

Auth

POST /api/auth/register

POST /api/auth/login

GET /api/auth/profile

Products

GET /api/products

GET /api/products/:id

POST /api/products

PUT /api/products/:id

DELETE /api/products/:id

Cart

GET /api/cart

POST /api/cart

PUT /api/cart/:id

DELETE /api/cart/:id

Orders

POST /api/orders

GET /api/orders

PUT /api/orders/:id

Admin

GET /api/admin/dashboard

GET /api/admin/users

DELETE /api/admin/users/:id

Document 06 — Implementation Plan

Phase 1

Project Setup

- Create React App
 - Setup Express
 - Configure MySQL
 - GitHub Repository
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Phase 2

Database Design

- Create Schema
 - Create Relationships
 - Seed Data
-

Phase 3

Authentication

- Register
 - Login
 - JWT Middleware
 - Protected Routes
-

Phase 4

Product Module

- Product CRUD
- Search
- Filters

- Pagination
-

Phase 5

Cart Module

- Add To Cart
 - Update Quantity
 - Remove Product
-

Phase 6

Order Module

- Checkout
 - Order Placement
 - Order History
-

Phase 7

Admin Dashboard

- Products
 - Orders
 - Users
-

Phase 8

Responsive Design

- Mobile
 - Tablet
 - Desktop
-

Phase 9

Testing

- Authentication Testing
- Product Testing
- Cart Testing
- Order Testing

Phase 10

Deployment

Frontend: Vercel

Backend: Render

Database: MySQL

Resume Description

Developed a production-style full-stack electronics e-commerce platform using React.js, Tailwind CSS, Node.js, Express.js, and MySQL. Implemented JWT authentication, role-based access control, product management, shopping cart functionality, order processing, inventory tracking, RESTful APIs, and an admin dashboard. Designed a scalable MVC architecture with responsive UI and optimized database relationships suitable for real-world deployment.